Schema Anomalies

Esme Basnet

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material.

Issues with Un-Normalized Data

Atomicity Anomaly

Update Anomaly

Insertion Anomaly

Deletion Anomaly

Atomicity Anomaly

Atomicity Anomaly - cells contain multiple sub-values packed together. If we ever need any of those sub-values then the DB will not help us (much) in finding and using them.

Atomicity Anomaly - Bank Customers

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ID	Name	Account IDs
1010	Joe Flint, Jane Flint	2590, 2013
5432	Jessica Bailey Leslie Marcus	992, 93292, 54

Atomicity Anomaly - Bank Customers

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1010	Joe Flint, Jane Flint	2590, 2013
5432	Jessica Bailey Leslie Marcus	992, 93292, 54

advice: avoid doing this by explicitly modeling the 1:n relationships both in the conceptual design and logical design.

Insertion Anomaly

Table is designed such that we cannot know/record some data until later

Insertion Anomaly - Superbowl Results

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ID	Date	Title	Winner	score	Loser	score	MVP	Site	City	State
1	Jan 15 1967	I	Green Bay Packers	35	Kansas City Chiefs	10	Bart Starr+	Memorial Coliseum	Los Angeles	California
2	Jan 14 1968	II	Green Bay Packers	33	Oakland Raiders	14	Bart Starr+	Miami Orange Bowl	Miami	Florida
55	Feb 7, 2021	LV	NULL	NULL	NULL	NULL	NULL	Raymond James Stadium	Tampa	Florida

Insertion Anomaly - Superbowl Results

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55	Feb 7, 2021	LV	NULL	NULL	NULL	NULL	NULL	Raymond James Stadium	Tampa	Florida

advice: separate the “known later” dynamic data into a separate table.

Update Anomaly

Concepts stored redundantly. So an update to one row might be missed in others

ID	Date	Title	Winner	score	Loser	score	MVP	Site	City	State
1	Jan 15 1967	I	Green Bay Packers	35	Kansas City Chiefs	10	Bart Starr+	Memorial Coliseum	Los Angeles	California
2	Jan 14 1968	II	Green Bay Packers	33	Oakland Raiders	14	Bart Starr+	Miami Orange Bowl	Miami	Florida
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Superbowls - Functional Dependencies

ID	Date	Title	Winner	score	Loser	score	MVP	Site	City	State
1	Jan 15 1967	I (1)	Green Bay Packers	35	Kansas City Chiefs	10	Bart Starr+	Memorial Coliseum	Los Angeles	California
2	Jan 14 1968	II (2)	Green Bay Packers	33	Oakland Raiders	14	Bart Starr+	Orange Bowl	Miami	Florida
3	Jan 12 1969	III (3)	New York Jets	16	Baltimore Colts	7	Joe Namath+	Orange Bowl	Miami	Florida

Superbowls - Troublesome Functional Dependencies

ID	Date	Title	Winner	score	Loser	score	MVP	Site	City	State
1	Jan 15 1967	I (1)	Green Bay Packers	35	Kansas City Chiefs	10	Bart Starr+	Memorial Coliseum	Los Angeles	California
2	Jan 14 1968	II (2)	Green Bay Packers	33	Oakland Raiders	14	Bart Starr+	Miami Orange Bowl	Miami	Florida
3	Jan 12 1969	III (3)	New York Jets	16	Baltimore Colts	7	Joe Namath+	Orange Bowl	Miami	Florida

Update Anomaly

update to one row might be missed in others

ID	Date	Title	Winner	score	Loser	score	MVP	Site	City	State
1	Jan 15 1967	I (1)	Green Bay Packers	35	Kansas City Chiefs	10	Bart Starr+	Memorial Coliseum	Los Angeles	California
2	Jan 14 1968	II (2)	Green Bay Packers	33	Oakland Raiders	14	Bart Starr+	Miami Orange Bowl	Miami	Florida
3	Jan 12 1969	III (3)	New York Jets	16	Baltimore Colts	7	Joe Namath+	Orange Bowl	Miami	Florida

advice: lift the troublesome functional dependency data into a separate table. In this case, create a “stadium” table.

Deletion Anomaly

If we delete a row, then we also delete other concepts (e.g., Los Angeles)

ID	Date	Title	Winner	score	Loser	score	MVP	Site	City	State
1	Jan 15 1967	I (1)	Green Bay Packers	35	Kansas City Chiefs	10	Bart Starr+	Memorial Coliseum	Los Angeles	California
2	Jan 14 1968	II (2)	Green Bay Packers	33	Oakland Raiders	14	Bart Starr+	Miami Orange Bowl	Miami	Florida
3	Jan 12 1969	III (3)	New York Jets	16	Baltimore Colts	7	Joe Namath+	Orange Bowl	Miami	Florida

advice: model important concepts with their own table(s). “stadium”, “city”, “state”, “player”, etc.

Normalization of Superbowl Data

Superbowls

ID	Date	Title	Winner	Score	Loser	Score	MVP	Site_ID
1	Jan 15 1967	I (1)	Green Bay Packers	35	Kansas City Chiefs	10	Bart Starr+	1
2	Jan 14 1968	II (2)	Green Bay Packers	33	Oakland Raiders	14	Bart Starr+	2
...	...	...	...					

Stadiums

ID	Name	City	State
1	Memorial Coliseum	Los Angeles	CA
2	Miami Orange Bowl	Miami	FL
...	...	...	...

Note: another "troublesome" functional dependency

Stadiums

ID	Name	City	State
1	Memorial Coliseum	Los Angeles	CA
2	Miami Orange Bowl	Miami	FL
...	...	...	...

Un-normalized data is very common!

E.g.: data scientists often work with single-table data sets

Does that mean that it is a good thing?   Or at least not a bad thing?

...depends on what you are trying to accomplish!

What are your priorities?

> Simplicity? (fewer tables, better read performance)
> Efficiency? (eliminate redundancies, better write performance)

Are Updates, Insertions and Deletions really a concern for your system?

Summary: Issues with un-Normalized Data

Atomicity Anomaly - cells contain multiple sub-values packed together

Update Anomaly - update to a row needs might fail to happen to all affected rows

Insertion Anomaly - we only record some facts when an entire row is inserted

Deletion Anomaly - deleting a row might delete knowledge of other entities

De-normalization: making a database less-normalized to increase read performance at the expense of redundancy and write performance.